

SAMEER PANDEY

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EDUCATION

Dr. D. Y. Patil Institute of Technology

Bachelor of Engineering in Artificial Intelligence & Data Science

Second Year | CGPA: 8.71 (First Year)

Pune, India

Sep 2025 – Present

SKILLS SUMMARY

Languages & Tools: Python, FastAPI, REST API, PostgreSQL, Git, Docker, Redis, SEO

Frameworks & Platforms: React, TypeScript, Nginx, Docker Compose, WebSocket, Redis Pub/Sub

Technical Areas: AI/ML integration, full-stack development, system design, RL environments, Data Structures & Algorithms (practicing)

Languages: English, Hindi

PROJECTS

CaloRupee – AI Budget Meal Planner

2026

Tech Stack: Python, FastAPI, React, TypeScript, PostgreSQL, AI Integration

- Independently built and deployed a full-stack web app helping students and budget-conscious users meet daily calorie and protein targets within a fixed rupee budget.
- Engineered AI-powered personalised 4-meal day plans with per-meal cost breakdown, accepting user inputs for location, ingredients, dietary preferences, and nutritional goals.
- Integrated two AI providers with automatic failover, ensuring 100% app availability even when one service experiences downtime.
- Built an interactive dashboard tracking meals checked off, calories consumed, protein intake, and money spent in real time, with a pre-built arsenal of 45 meals.

VisionLink – Real-Time Emotion Detection System

Tech Stack: FastAPI, React, PostgreSQL, Redis Pub/Sub, Docker Compose, Nginx, MediaPipe

- Built a real-time emotion detection system capturing video frames and classifying 6 emotions (Happy, Sad, Angry, Surprise, Wink, Neutral) using MediaPipe Face Mesh with a floating emoji HUD overlay.
- Architected a scalable backend using FastAPI and Redis Pub/Sub, enabling multiple workers to broadcast processed frames to unlimited WebSocket clients simultaneously.
- Delivered production-ready features including API key authentication, cursor-based pagination for detection history, background DB writes with error logging, and BuildKit-optimized Docker builds.
- Orchestrated full-stack system via Docker Compose: React frontend with Framer Motion animations, FastAPI backend, PostgreSQL for detection history, and Nginx as reverse proxy.

NutriSync – Reinforcement Learning Meal Planning Environment

Tech Stack: Python, OpenAI Gym, Groq/LLaMA, Gradio

- Designed an OpenEnv-compatible RL gym where an agent plans 4 sequential meals (Breakfast, Lunch, Dinner, Snack) to satisfy nutritional and budget constraints, scoring in [0, 1].
- Implemented a non-exploitable reward system using additive scoring with multiplicative penalties and hard kill-switches (e.g., allergen use or budget breach instantly zeros the score).
- Created three difficulty tiers – Easy (omnivore, ₹250), Medium (vegetarian, ₹200), Hard (vegan, ₹160) – with stricter calorie, protein, glycemic load, and availability constraints.
- Built a 50-ingredient Indian food database covering 10 clusters with cooking-method nutritional modifiers (frying +45% calories, fermenting boosts Vitamin C) and a Gradio UI for live exploration.

Package Scanner & Brand Trust Rating App

Sep 2026

Tech Stack: Mobile App (Android), Backend REST API, Public Web Dashboard

- Built a mobile app that lets users scan product packaging and aggregates scans per brand to compute a trust/quality rating after 3–4 scans.
- Published aggregated brand ratings to a public web dashboard so consumers can check a brand's rating before purchasing.

AWARDS & ACHIEVEMENTS

1st Place – Impromptu Hackathon (3-Hour Sprint)

March 2026

- Secured first place in a 3-hour sprint hackathon by building a full-stack college canteen web app featuring real-time menu updates, admin panel, and payment gateway integration.

CERTIFICATIONS & COURSEWORK

Relevant coursework and self-study: Artificial Intelligence, Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems